

Development of a Laser-Based Optical Sensor for Microplastic Monitoring

Plan of Action (Synopsis)

The proposed system detects microplastics in water using the principle of optical light attenuation and scattering. A water sample is passed through a light-proof (dark) flow chamber to eliminate interference from ambient light. A 650 nm laser diode is positioned on one side of the chamber to produce a narrow, stable beam through the flowing water.

On the opposite side, a photodiode measures the transmitted light intensity, while additional photodiodes/LDRs placed at angles (e.g., 45° or 90°) detect light scattered by suspended particles. When microplastic particles pass through the laser beam, they absorb and scatter part of the light, causing measurable changes in the sensor outputs.

The analog signals from the sensors are amplified and converted into digital values using a microcontroller's ADC (e.g., Arduino or ESP32). Signal-processing algorithms are then used to identify significant fluctuations corresponding to particle passage while filtering out noise caused by water turbulence or bubbles.

The processed data is displayed on an LCD or transmitted to a computer/mobile application, providing an estimate of the presence and relative concentration of microplastics. The system is intended to be low-cost, portable, non-destructive, and suitable for preliminary field screening of freshwater sources before laboratory confirmation.

Workflow

- 1 Collect water sample.
- 2 Pump the sample through a dark optical chamber.
- 3 Illuminate the sample using a 650 nm laser diode.
- 4 Measure transmitted and scattered light using photodiodes/LDRs.
- 5 Amplify and digitize sensor signals.
- 6 Process the data to detect attenuation/scattering patterns.
- 7 Display or log the estimated microplastic concentration.